Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/08/2007
PI Name:	Goldberg, Alfred L Ph.D.		
Project Title:	The Activation of Protein Breakdown in Muscle Upon Unloading and Possible Countermeasures		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Muscle Alterations and Atrophy Team		
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Human Research Program Risks:	None		
Space Biology Element:	None		
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
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Task Description:	The rapid loss of muscle mass that occurs in astronauts in space due to muscle unloading and in patients with many systemic diseases results primarily from accelerated degradation of muscle proteins. This enhancement of protein breakdown is mainly due to activation of the Ub-proteasome pathway. Our major goal is to clarify the biochemical basis for the activation of this proteolytic pathway and thus to develop pharmacological agents that reduce this excessive degradation and retard muscle wasting. Recently, we identified a set of genes ("atrogenes") whose expression increases or decreases coordinately when muscles atrophy. In order to achieve a fuller understanding of the atrophy process and to develop novel inhibitors of the atrophy process, we plan to further study this transcriptional program and its regulation, especially after disuse. Of particular importance was the recent finding that the two genes induced most dramatically in atrophying muscles are the muscle-specific ubiquitin ligases (E3s), atrogin-1 (MAFbx) and MuRF1. If either of these enzymes is knocked out, the extent of muscle wasting is reduced. These Ub-ligases thus are very attractive therapeutic targets. To fully understand the initiation of the atrophy process, and to develop rational countermeasures, we are also studying the signal transduction systems that activate transcription of these genes in simple models of muscle atrophy in cultured myotubes. We recently found that the key factor in muscle hypertrophy, IGF-1, rapidly suppresses the expression of atrogin-1 and MuRF1 and prevents their induction by glucocorticoids. These effects of IGF-1 appear to be mediated by the PI3-kinase-AKT pathway, which probably inactivates one of the Forkhead transcription factors. Our primary goals will be to identify the precise steps in this kinase cascade and the key transcription steps through which disuse and glucocorticoids activate and IGF-1 inhibits expression of atrogin-1 and MuRF1. These studies should identify novel therapeutic targets (e.g. key kinases) or Foxo-regulatory factors whose inhibition blocks atrogin-1 and MuRF1 induction. Using these enzymes and "reporter gene" constructs, we shall screen libraries of small molecules for agents that prevent induction of atrogin-1 and MuRF1. Inhibiting the expression of these key ligases represents an exciting new therapeutic approach to prevent muscle wasting in space personnel and in diverse disease states. In addition, we are continuing to test whether inhibitors of proteasomes, both proved ones and the inhibitor (Velcade) now used in cancer therapy by partially retarding overall proteolysis may be useful in retarding muscle wasting.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	If successful, this research may yield novel therapies to prevent or treat the marked wasting of muscle seen in various bed-ridden patients (i.e. the elderly) and ones with various systemic diseases (including cardiac failure, cancer, sepsis, renal failure, AIDS).
Task Progress:	A) Transcriptional Changes in Rapidly Atrophying and Atrophied Muscles. Muscle atrophy occurs upon unloading and systemically in fasting and many catabolic diseases. We have identified a common set of genes (“atrogenes”), whose expression is coordinately induced or suppressed in muscle during these various wasting states. To determine whether this transcriptional program also functions during atrophy resulting from inactivity and if atrogene expression correlates with the rate of weight loss, we analyzed changes in gastrocnemius mRNA following denervation or disuse induced by spinal isolation. After these procedures, the loss of muscle weight was initially rapid and then slowed. At 3 days, 78% of the atrogenes identified during catabolic states were induced or repressed, especially the key ubiquitin-ligases, atrogin-1 and MuRF1, but at 14 days when atrophy has slowed, the expression of 92% of the atrogenes returned to basal levels, including all atrogenes related to protein breakdown. Subsequently, expression of a few atrogenes remained high in the disused muscle, including mRNA for the transcription factor, Foxo1, which may be important in maintenance of the “atrophied” state. Thus, 1) the atrophy in systemic diseases and following disuse share similar transcriptional adaptations. 2) Atrophy induced by disuse proceeds through multiple phases: the rapidly “atrophying” phase (when atrogene expression is pronounced) and “atrophied” muscles. 3) Because expression of atrogin-1 and MuRF1 appear responsible for the rapid muscle weight loss, these mRNAs can serve as useful biomarkers for evaluating countermeasures. B) The Exercise-Induced Coactivator PGC-1 Inhibits Muscle Atrophy by Blocking FoxO3. Our studies indicated that in atrophy induced by unloading, fasting, and systemic diseases, the FoxO family of transcription factors plays a critical role in activating protein breakdown. When activated, FoxO3 itself triggers expression of atrogin-1 and MuRF1 and causes profound atrophy. Physical exercise also influences fiber-type composition as well as fiber size in muscle. Others have shown that increased activity stimulates production of the transcriptional coactivator PGC-1alpha, which promotes switching from glycolytic towards more oxidative fibers and production of mitochondria. To understand how exercise might retard atrophy, we investigated whether PGC-1alpha is also important in the regulation of muscle size. In atrophying rodent muscles, the atrophying fibers showed markedly reduced expression of PGC-1alpha. In transgenic mice overexpressing PGC-1alpha, denervation caused little or no loss of fiber diameter, and atrogin-1 and MuRF1 induction was blocked. Also, transfection of PGC-1alpha into adult muscle fibers reduced the capacity of FoxO3 to cause fiber atrophy. Thus, PGC-1alpha and FoxO3 are important opposing factors in determining muscle mass. The production of PGC-1alpha upon exercise seems to reduce the catabolic effects of FoxO3, while the fall of PGC-1alpha with inactivity and catabolic states contributes to the FoxO-dependent loss of muscle protein. C) Inhibition of Muscle Atrophy by Administrating the Proteasome Inhibitor, Velcade. Because we demonstrated a key role of the proteasome in catalyzing the loss of muscle protein during atrophy, we investigated the capacity of Velcade (PS341), the proteasome inhibitor now approved by the FDA for cancer treatment to retard muscle atrophy. Following a section of the sciatic nerve in mice, clear atrophy of the gastrocnemius was evident by seven days. Injection of Velcade every other day reduced the weight loss of the muscles by 40% without any obvious toxicity. Studies are in progress to optimize the effective dose and to learn whether the inhibitor affects other forms of atrophy and whether these muscles also have greater functional capacity than those in untreated animals.
Bibliography Type:	Description: (Last Updated: 09/01/2017)

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Awards	Goldberg AL. "Centennial Lecturer, Biochem Society, January 2006." Jan-2006
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